

Axiom Shopping Assistant

CIS 5500 final delivery

- Search, compare, and rank Amazon products
- Review-backed product detail and alternatives
- Analytics views for categories, brands, and review trends

Datasets

Raw CSVs to normalized PostgreSQL tables

- Products: 1,426,337 raw rows
- Reviews: 701,528 raw rows; 11,927 linked in current corpus
- Categories: 248 lookup rows
- Cleaner canonicalizes brands to 26,484 entities

Products



1,426

Reviews



701,5

Categories

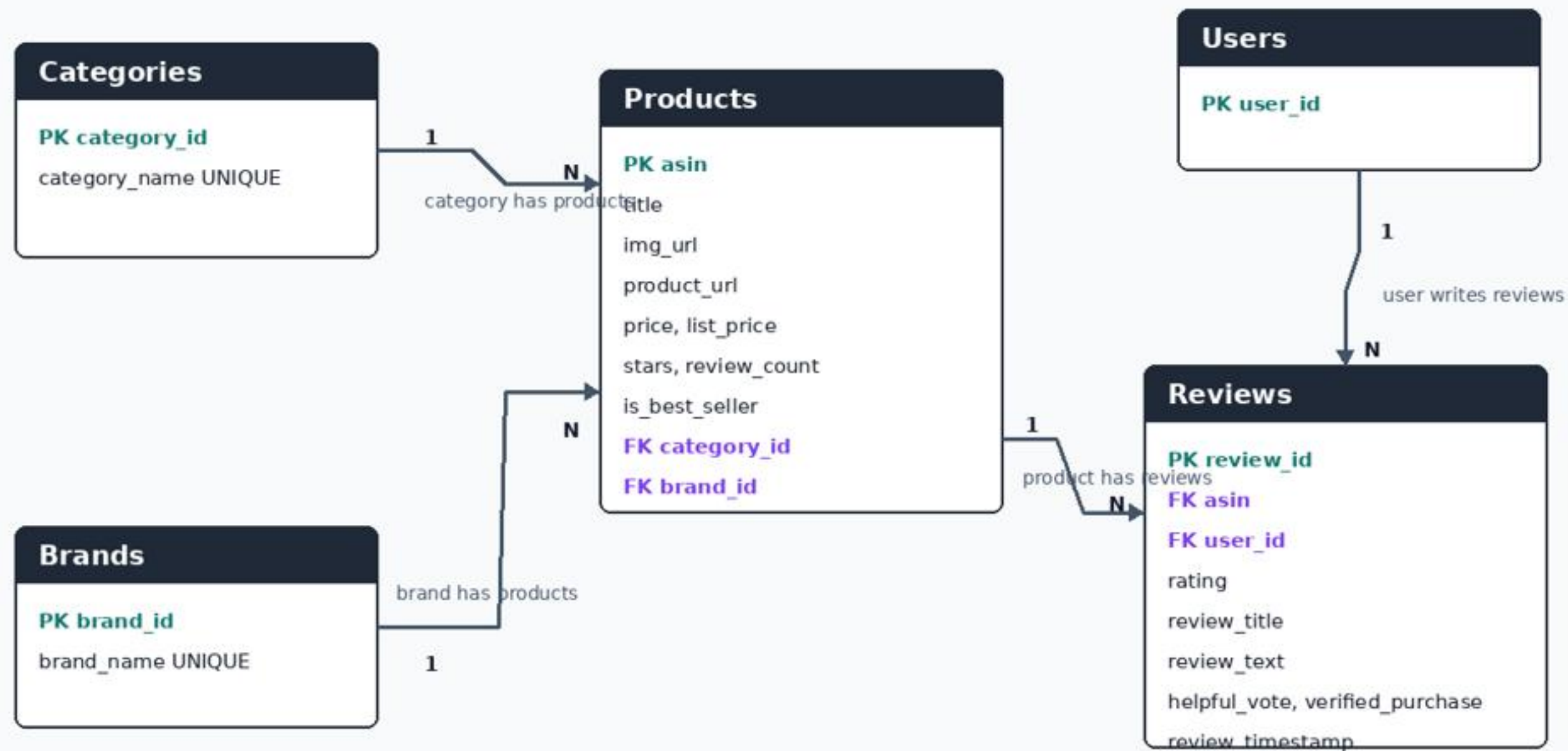
248

ER Diagram

Five relations, four foreign-key links

Axiom Shopping Assistant ER Diagram

Five normalized relations with product, category, brand, user, and review links



Schema and 3NF

Primary-key determinants; no transitive dependencies

- Categories and Brands use ID primary keys plus unique names
- Products are determined by ASIN and reference category and brand
- Reviews are determined by review_id and reference product and user
- Users store only user_id, so no non-key dependencies exist

Complex Query: Brand Performance

Verified-review analytics per brand

- Original path aggregates product stats and review stats, then joins
- Optimized path reads mv_brand_performance
- Median timing: 2849.7 ms to 0.0 ms

Complex Query: Review Trend

Monthly category quality signal

- Groups reviews by month for a selected category
- Route cache handles repeat analytics hits
- Median timing: 34.8 ms to 34.6 ms

Complex Query: Top Value and Trending

Category averages plus recent review windows

- Top value reads mv_value_score_components
- Top-value median: 444.4 ms to 0.8 ms
- Trending median: 22.6 ms to 4.2 ms

Complex Query: Value Rankings

Weighted product score

- Normalizes rating, review count, price efficiency, and recent reviews
- Optimized path uses indexed default-score components
- Median timing: 9223.5 ms to 1.7 ms

Technical Challenges

What changed for final delivery

- Brand entity resolution from noisy product titles
- ASIN mismatch between review and product sources
- Materialized views plus refresh maintenance
- Input validation without changing public API contracts

Demo and Extra Credit

Final walkthrough

- Home/search to product detail
- Cart savings
- Analytics charts
- Value rankings sliders
- Deployment URL: add after hosting